

Future Prospects in Gravitational Waves: From Testing Fundamental Physics to Instruments beyond LIGO

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Thesis Defense

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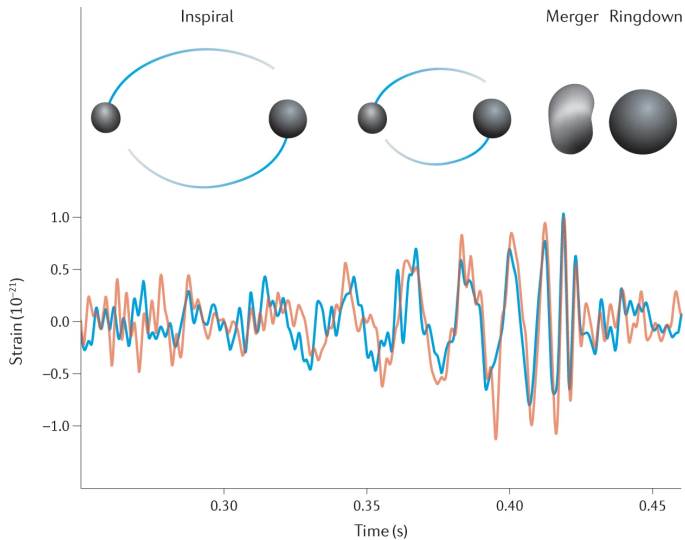
Supervisor: Yanbei Chen



Outline

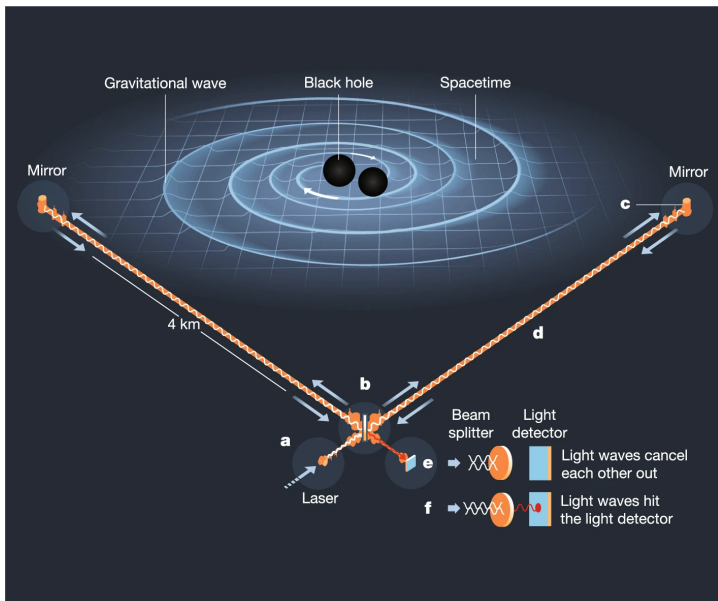
- 1 Introduction
 - Science with Gravitational Waves
 - Features of GR & Beyond GR Waveforms
- 2 Part A : Searching for Nonviolent Nonlocality in the Gravitational Waves
 - Motivation for Nonviolent Nonlocality
 - Waveforms in Nonviolent Nonlocality
 - Estimating the Upper Bound on Metric Fluctuations
- 3 Part B : Geometric Description of Tests of GR
 - Geometry of Waveform Deviations
 - Generic Behavior of Parameterized Tests
 - Multiparameter Tests with Singular Value Decomposition
- 4 Part C : High Frequency Gravitational Wave Detection
- 5 Conclusion

Introduction



- Image of first detection of gravitational waves in 2015.

[LIGO+ 2016, Bailes+ 2021]



[Miller+ 2019]

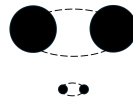
Current & Future Science Goals

- 1 Astrophysics of compact objects: formation scenarios and exotic systems
- 2 Nuclear physics: tidal deformability and neutron-star equation of state
- 3 Multimessenger astrophysics: electromagnetic counterparts and kilonova
- 4 Cosmology: independently measuring expansion of universe
- 5 Stochastic background: unresolved background of gravitational waves
- 6 Lensing signatures
- 7 Fundamental physics: testing consistency with general relativity

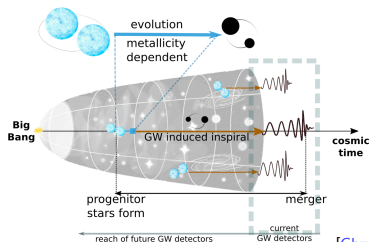
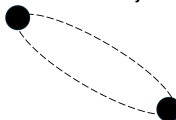
Large Asymmetric Spins



Large or Small Masses



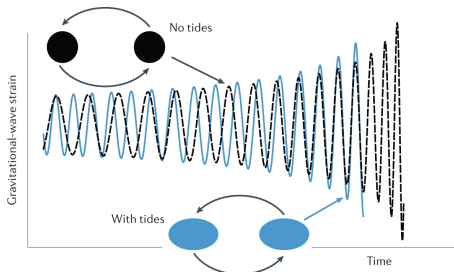
Eccentricity



[Chruslinska 2022]

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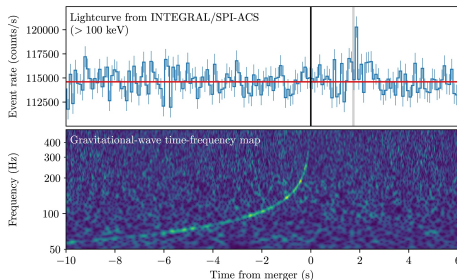
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[Carson 2020]

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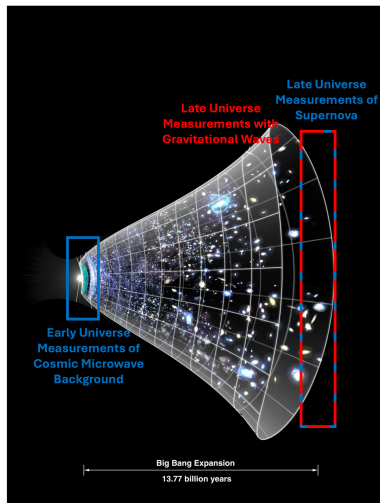
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[LIGO+ 2017]

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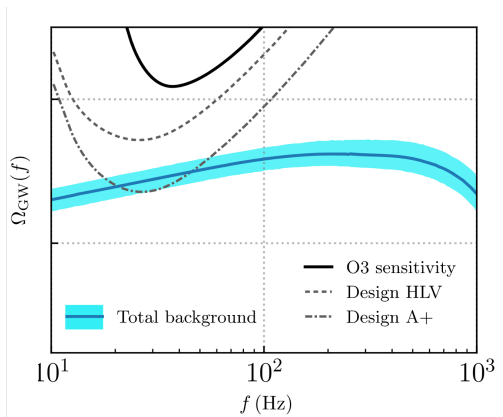
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[NASA / WMAP Science Team]

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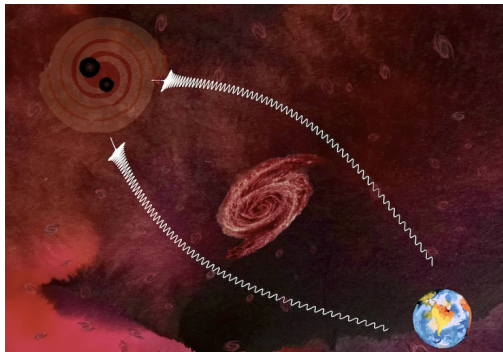
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[LIGO+ 2023]

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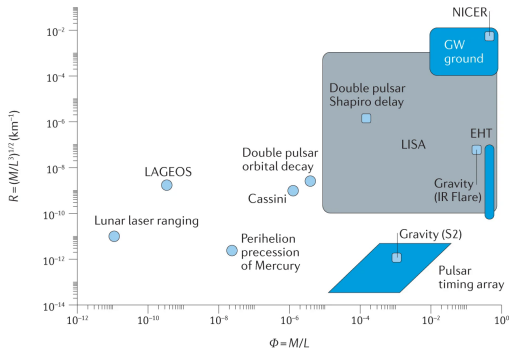
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[Jana+ 2023]

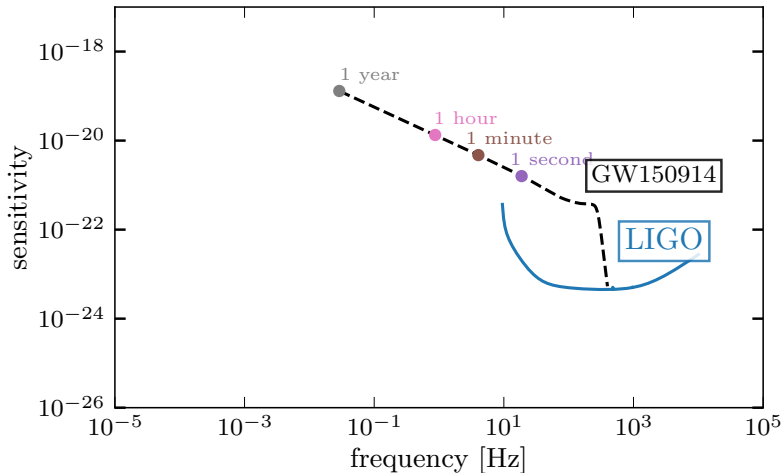
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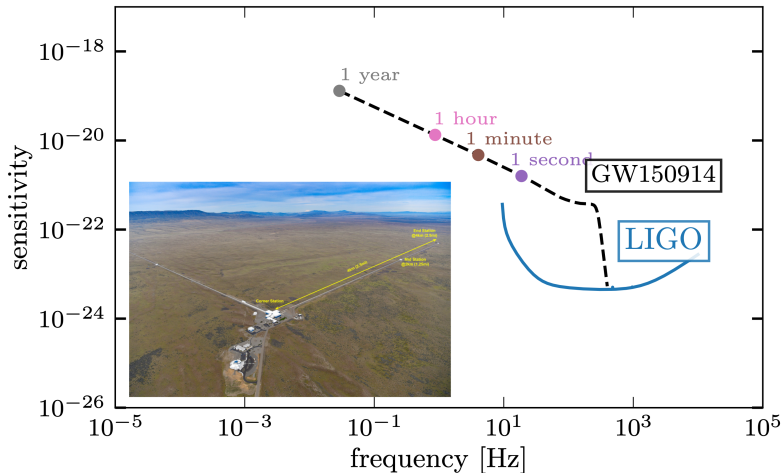


[Yunes+ 2016, Bailes+ 2021]

Current & Future Detectors

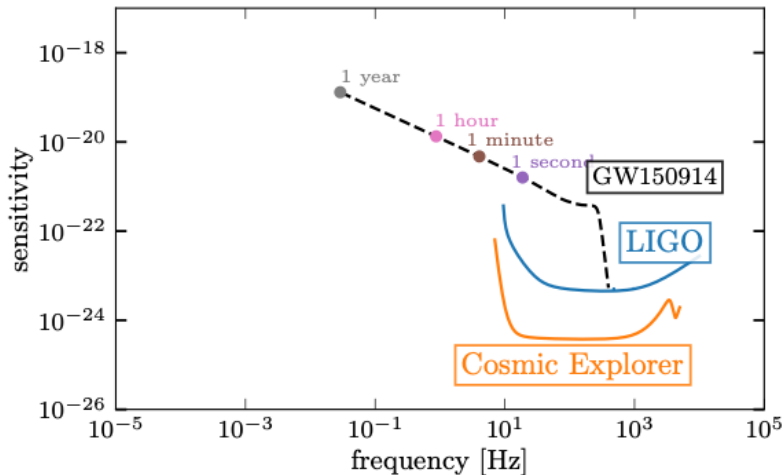


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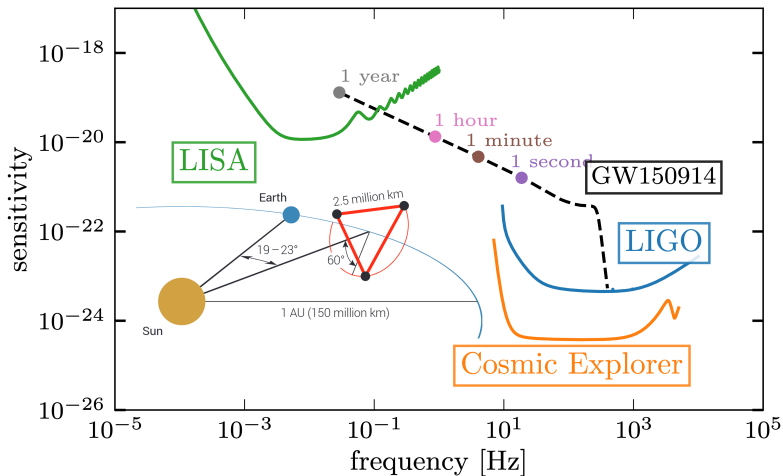


[LIGO Lab]

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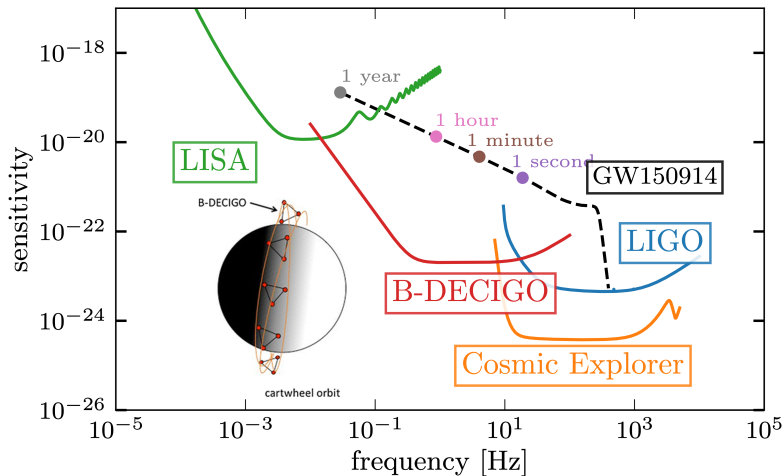


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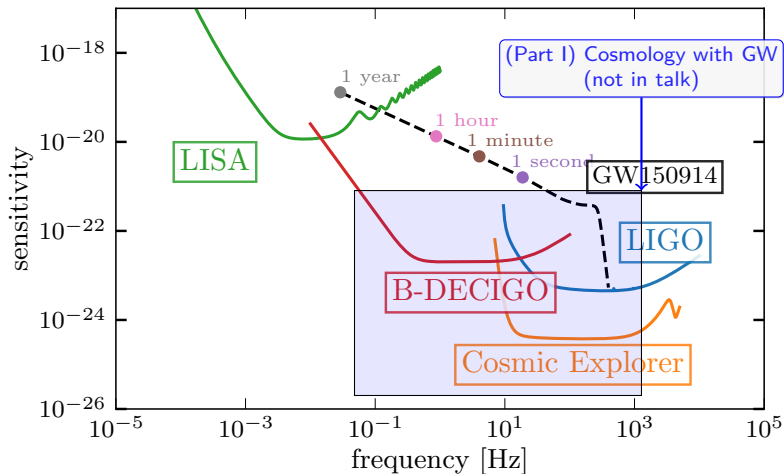
[LISA Collaboration 2017]

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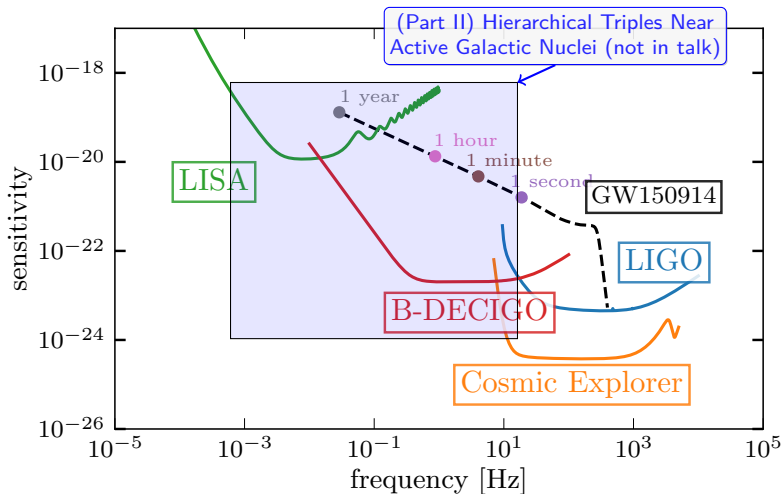


[DECIGO 2017]

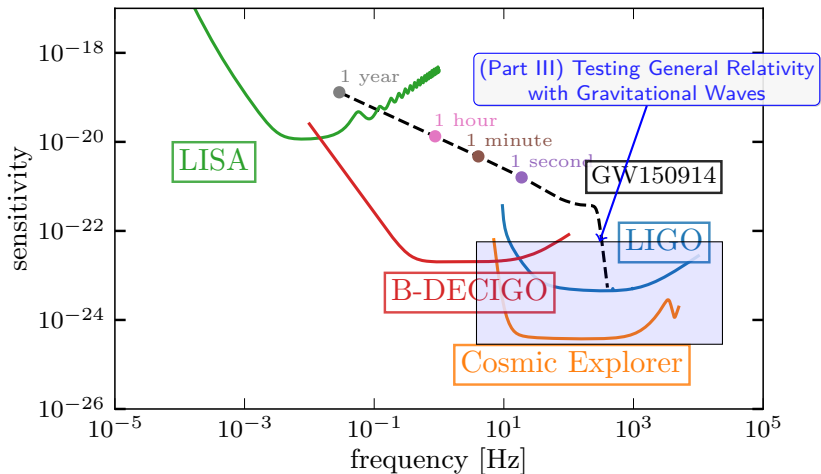
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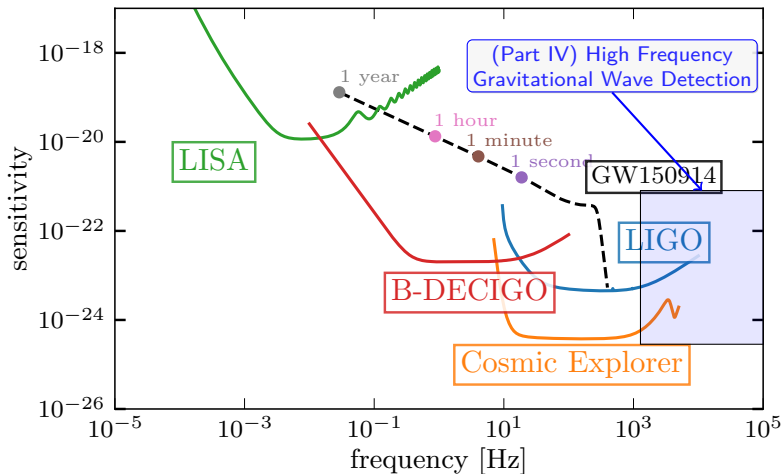
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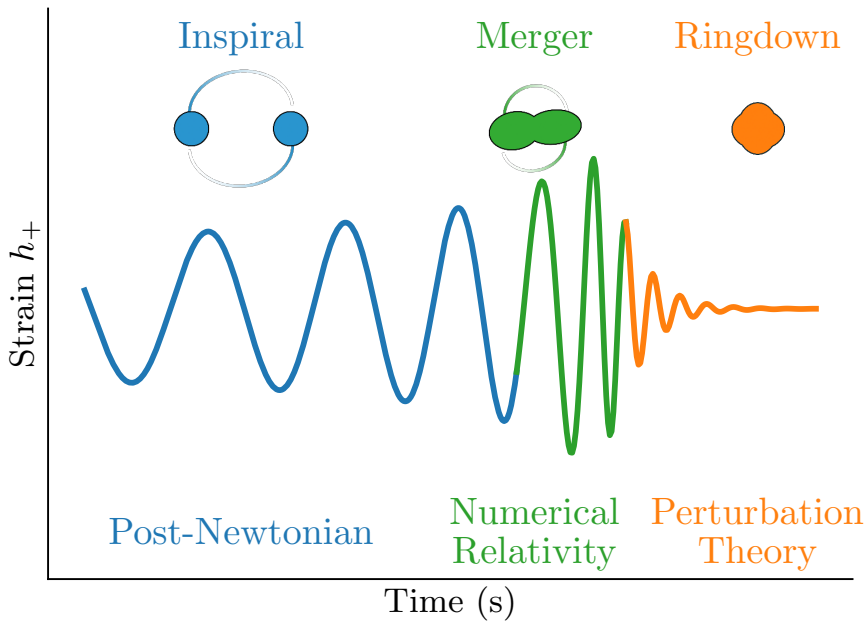


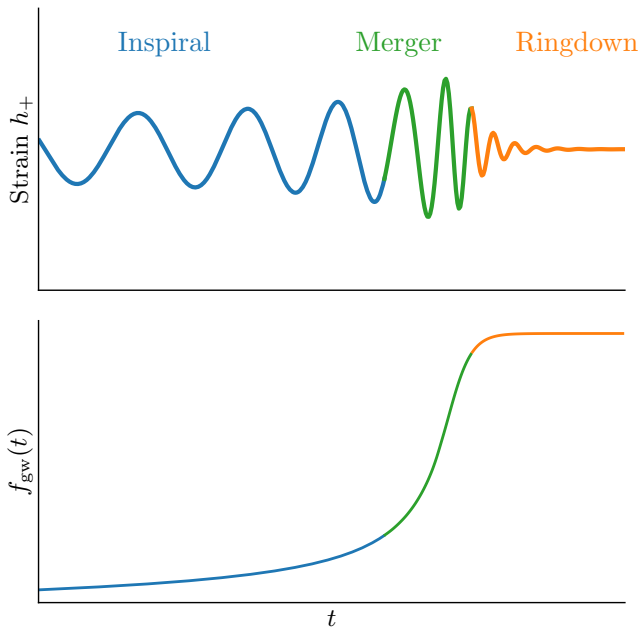
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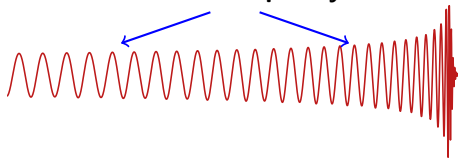






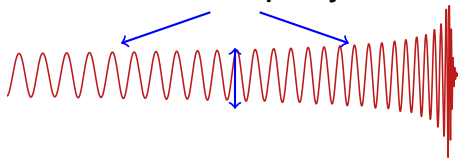
Gravitational Waves Encode Intrinsic Binary Physics

masses inferred via **frequency evolution**



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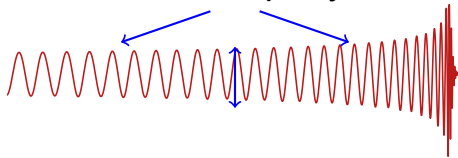
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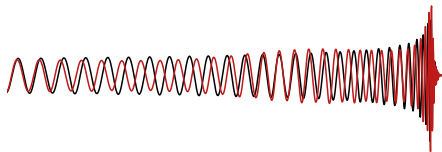
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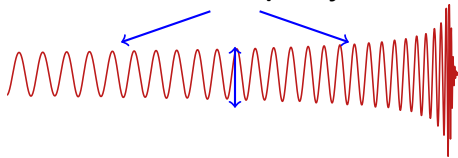
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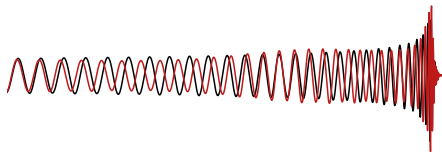
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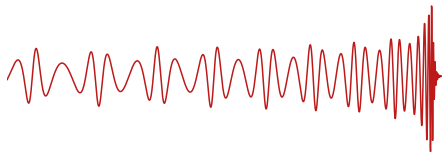
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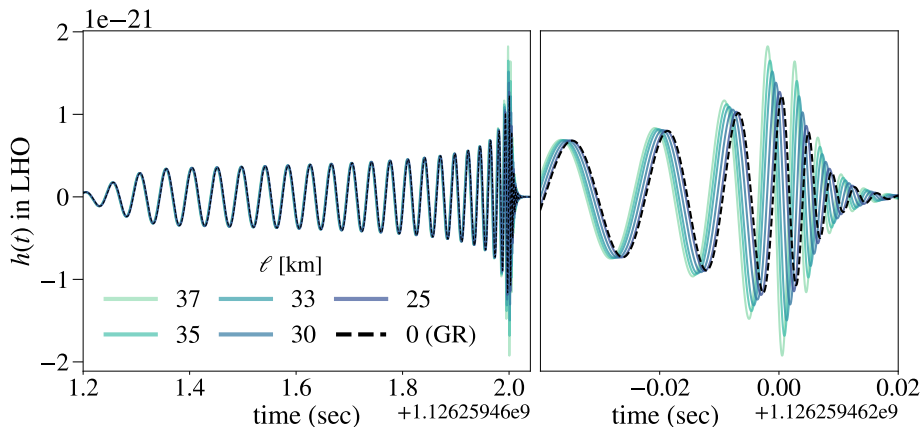


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eccentricity also manifests in
amplitude and **phase** modulations

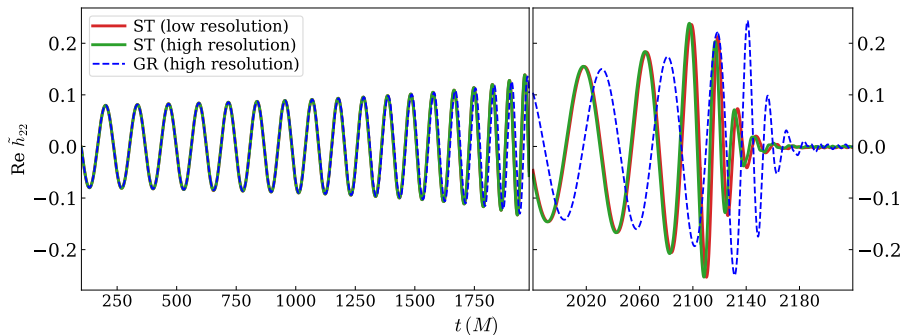
Beyond GR Waveforms



- Beyond GR simulation of dynamical Chern-Simons [Okounkova+ 2023]

Beyond GR Waveforms

$$m_{\text{BH}}^{\text{J}} = 5.7 M_{\odot}, m_{\text{NS}}^{\text{J}} = 1.5 M_{\odot}, \chi_z^{\text{BH}} = -0.19, \chi_z^{\text{NS}} = 0, \Lambda_2^{\text{GR}} = 131.1$$



- Beyond GR simulation of scalar tensor [Ma+ 2023]

Parameterized Tests of General Relativity

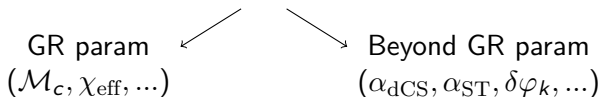
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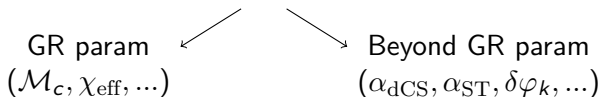
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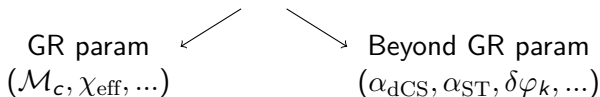
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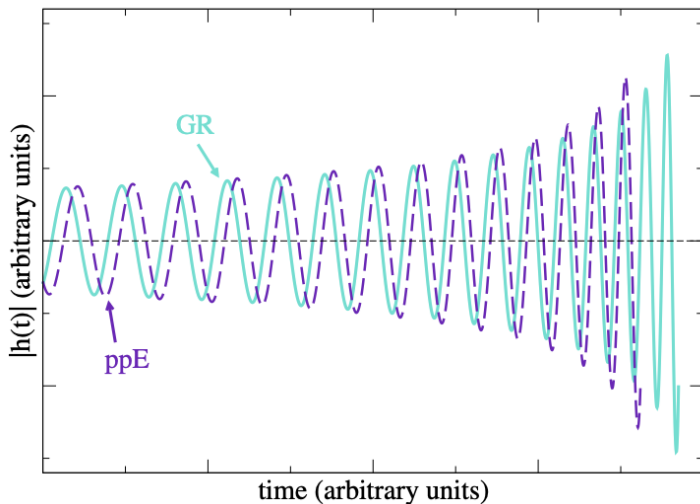
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where each of these $\delta\varphi_k \leftrightarrow$ deviation of $(M/r)^k$ away from GR

PPE in Time Domain

- These ppE deviations in time domain look like [Carson+ 2020]



How do We Interpret Parameterized Constraints

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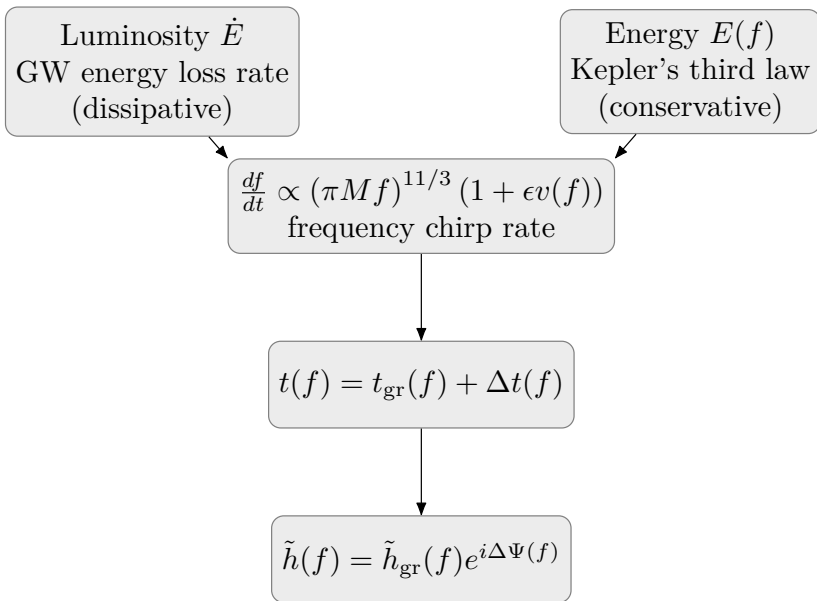
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- $\frac{df}{dE}$ is due to modified Kepler's third law, $\frac{dE}{dt}$ due to dissipative modifications.
- A beyond GR effect causes relative time delays $\Delta t(f)$ and then the stationary phase approximation for an adiabatic energy loss rate implies [Yunes+ 2009, Tahura+ 2019]

$$\Delta\Psi(f) = 2\pi \int df \Delta t(f)$$



Part A : Searching for Nonviolent Nonlocality in the Gravitational Waves

Possibilities for the Black Hole Unitarity Crisis

It is proposed that the following three statements cannot be true simultaneously [[Almheiri+ 2012](#)]

- 1 Hawking radiation is a pure state.
- 2 Infalling observer feels nothing unusual at the horizon.
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- ③ Hawking radiation comes from near the horizon.
 - Horizon structure of a black hole is changed, e.g. nonviolent nonlocality.

Nonviolent Nonlocality

- Nonviolent nonlocality is a proposal by Steve Giddings that posits that the information is transferred via soft modes in the black hole atmosphere [[Giddings 2012](#), [Giddings+ 2016](#)].
- These metric fluctuations have an extent to $\sim r_S$ in contrast to the fluctuations in a firewall with extent $l_p \ll r_S$
- We have background metric, and the quantum fluctuations modify it

$$g_{\mu\nu} = g_{\mu\nu}^{\text{schw}} + n_{\mu\nu}$$

Modifications to the Metric due to Quantum Structure

- One can construct the most general metric fluctuations [Regge+ 1957], but the dominant one in ingoing Eddington-Finkelstein coordinates is [Giddings+ 2016]

$$n_{vv} = \sum_{\ell m} f^{\ell m}(v, r) Y_{\ell m}(\phi, \theta)$$

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- We parameterize the random noise fluctuations as

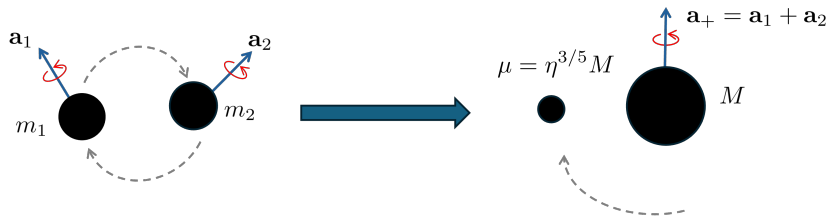
$$f^{\ell m}(v, r) = A \exp \left[-\frac{1}{2r_S^2} (r - r_S)^2 \right] n(t)$$

$$n(t) = \text{Colored gaussian noise; } \langle |n(t)| \rangle = 1$$

$$S_n(f) \propto \frac{1}{2f_Q} \exp[-|f|/f_Q]$$

Effective One-Body

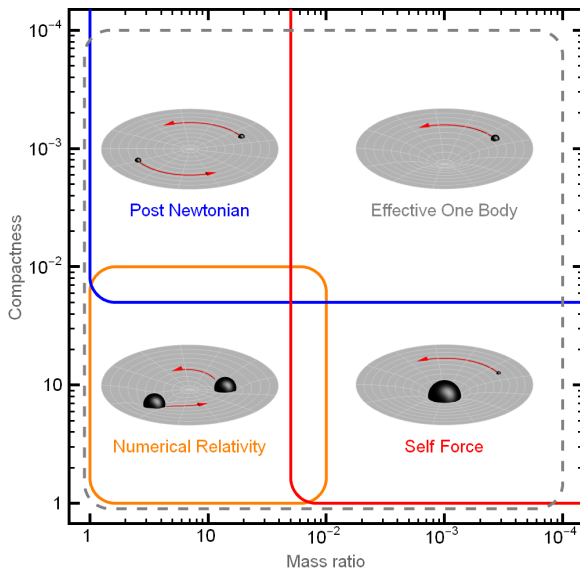
- The effective one-body formalism is an analytical approach to finding the motion and gravitational waves in GR [Buonanno+ 1998, 2000, Damour 2001].
- It resums independent information from (a) post-Newtonian theory and (b) black hole perturbation theory



Full GR two body problem

Point particle in effective spacetime
 η -deformed Kerr

Effective One-Body



Effective One-Body Details

- The effective one-body defines an effective metric

$$ds_{\text{eff}}^2 = -A(r)dt^2 + \frac{D(r)}{A(r)}dr^2 + r^2d\Omega^2$$

where if $\eta \rightarrow 0$ it approaches the Schwarzschild metric in the nonspinning case.

- One can find that solving the mass shell condition $p_\mu p_\nu g^{\mu\nu} = -1$ yields an effective Hamiltonian

$$\hat{H}_{\text{eff}} = \sqrt{A(r) \left(1 + \frac{p_\phi^2}{r^2} + \frac{A}{D} p_r^2 \right)}$$

- The particle follows the trajectory of the real Hamiltonian

$$\hat{H}_{\text{real}} = \frac{1}{\eta} \sqrt{1 + 2\eta \left(\hat{H}_{\text{eff}} - 1 \right) + \mathcal{O}(\vec{P}_{\text{COM}}^2)}$$

where the center of mass motion is ignored.

Perturbations to Hamilton's Equations

- Nonviolent nonlocality adds perturbations to black holes which changes the mass shell condition

$$p_\mu p_\nu (g_0^{\mu\nu} + h^{\mu\nu}) = -1$$

This causes the real Hamiltonian to be modified

$$\hat{H}_{\text{real}} = \hat{H}_{\text{real}}^0 + h_{\text{vv}}^{\ell m} \Delta \hat{H}_{\ell m}^{\text{real}} .$$

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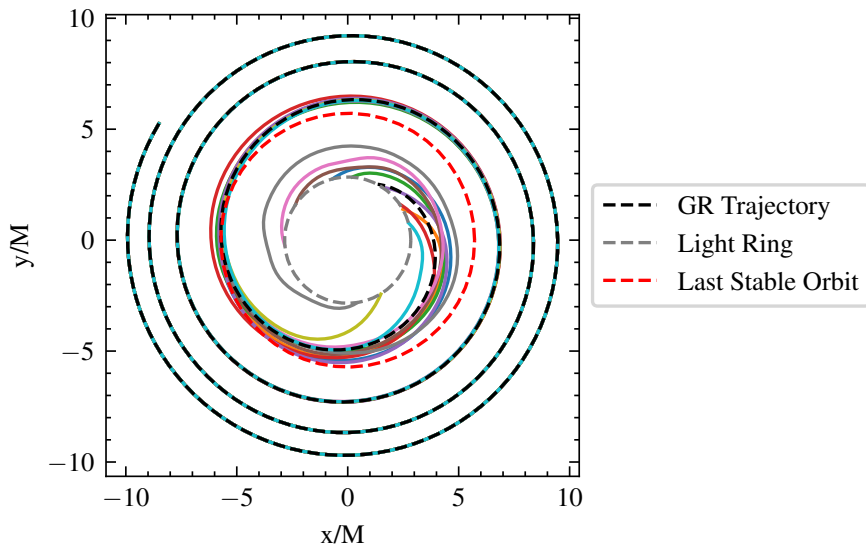
$$\hat{H}_{\text{real}} = \hat{H}_{\text{real}}^0 + h_{\text{vv}}^{\ell m} \Delta \hat{H}_{\ell m}^{\text{real}}.$$

- In the end, we integrate Hamilton's equations which have the form

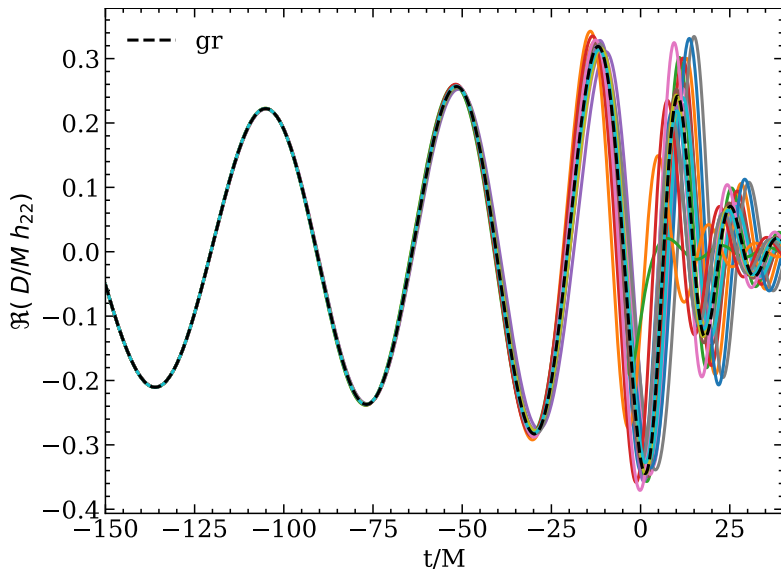
$$\begin{aligned} \frac{dq_i}{dt} &= \frac{\partial \hat{H}_{\text{real}}}{\partial p_i}, \\ \frac{dp_i}{dt} &= -\frac{\partial \hat{H}_{\text{real}}}{\partial q_i} + \mathcal{F}_i^{\text{rad}}, \end{aligned}$$

where $\mathcal{F}_i^{\text{rad}}$ are the nonconservative forces from radiation reaction.

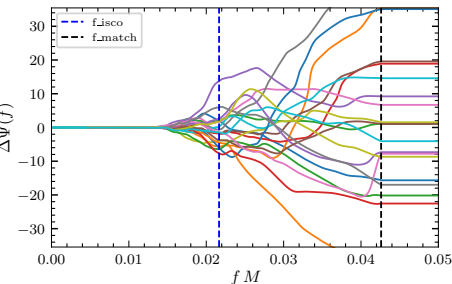
Inspiral with Quantum Fluctuations



Change in Gravitational Wave Strain

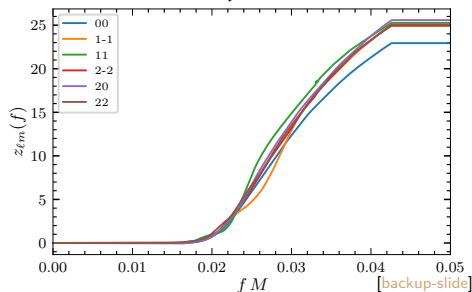


Random Dephasing in Frequency Domain



- Principal component analysis yields deviation that looks like $\Delta\Psi(f) = \zeta z(f)$
- $\zeta \sim \mathcal{N}(0, A_t)$ random variable for each event

PCA



Fisher Analysis

- We inject $\theta = (\zeta, \mathcal{M}_c, q, D_l, \iota, \psi, \alpha, \delta)$ from realistic astrophysical populations and compute the estimated variance with the Fisher matrix

$$\Gamma_{ij} = (\partial_{\theta_i} h | \partial_{\theta_j} h)$$

where $(.|.)$ is the standard noise weighted inner product.

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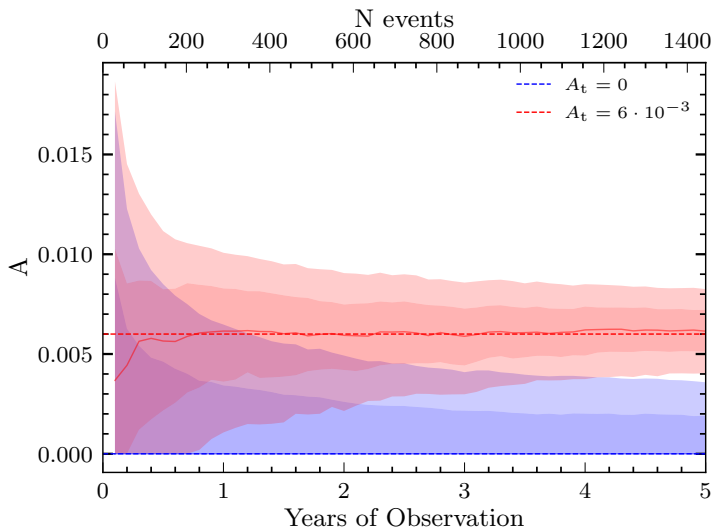
where $(.|.)$ is the standard noise weighted inner product.

- From this we calculate the marginalized likelihood $p(d_a | \zeta)$ for each event a . The likelihood for the hyper parameters is

$$p(d | \mu, \sigma) = \int d\zeta p(d | \zeta) p(\zeta | \mu, \sigma)$$

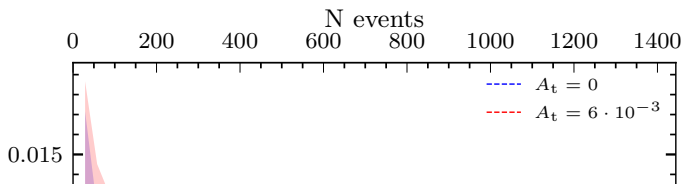
- From this, we compute the posterior on $A \equiv \sigma$ by combining many events d_a .

Constraints on Nonviolent Nonlocality with LIGO

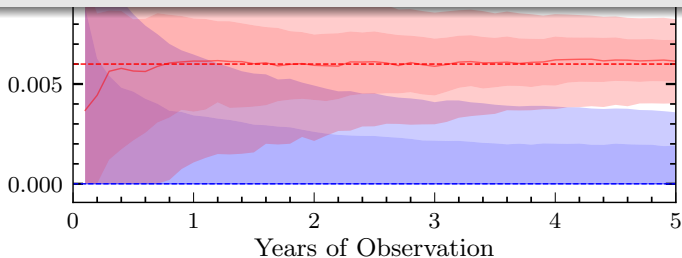


[backup-slide]

Constraints on Nonviolent Nonlocality with LIGO



Metric fluctuations can be constrained to be $A \lesssim 6 \times 10^{-3}$ with 5 years of O3 data.



[backup-slide]

Part B : Geometric Description of Tests of GR

Testing GR with Incorrect Signal Models

- Gravitational wave data is a combination of signal and noise

$$d = s + n$$
$$s = h_{\text{sig}}(\theta_{\text{tr}}, \alpha_{\text{tr}}) = h_{\text{gr}}(f; \theta_{\text{tr}}) e^{i\Delta\Psi_{\text{bgr}}}$$

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GR param

$(\mathcal{M}_c, \chi_{\text{eff}}, \dots)$

Beyond GR param

$(\alpha_{\text{dCS}}, \alpha_{\text{ST}}, \delta\varphi_k, \dots)$

The diagram illustrates the decomposition of the signal s into two parameter sets. The equation $s = h_{\text{sig}}(\theta_{\text{tr}}, \alpha_{\text{tr}}) = h_{\text{gr}}(f; \theta_{\text{tr}}) e^{i\Delta\Psi_{\text{bgr}}}$ is shown. Below it, two arrows originate from the parameters θ_{tr} and α_{tr} respectively. The arrow from θ_{tr} points to the 'GR param' box, which contains $(\mathcal{M}_c, \chi_{\text{eff}}, \dots)$. The arrow from α_{tr} points to the 'Beyond GR param' box, which contains $(\alpha_{\text{dCS}}, \alpha_{\text{ST}}, \delta\varphi_k, \dots)$.

Testing GR with Incorrect Signal Models

- Gravitational wave data is a combination of signal and noise

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Residual Signal

- Doing PE on $h_{\text{sig}}(\theta_{\text{tr}}, \alpha_{\text{tr}})$ with a GR waveform $h_{\text{gr}}(\theta)$, the residual signal

$$\begin{aligned}\Delta h &= h_{\text{sig}}(\theta_{\text{tr}}, \alpha_{\text{tr}}) - h_{\text{gr}}(\theta_{\text{tr}}) \\ &\approx i\alpha_{\text{tr}}\psi_{\alpha}h_{\text{gr}}(\theta_{\text{tr}})\end{aligned}$$

- GR parameters are biased due to systematic error [[Cutler+ 2007](#)]

$$\theta_{\text{meas}} \rightarrow \theta_{\text{tr}} + \Delta\theta_{\text{bias}}$$

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$$\Delta h_{\perp\text{gr}} = \Delta h - \Delta\theta_{\text{bias}}^i \partial_i h_{\text{gr}}$$

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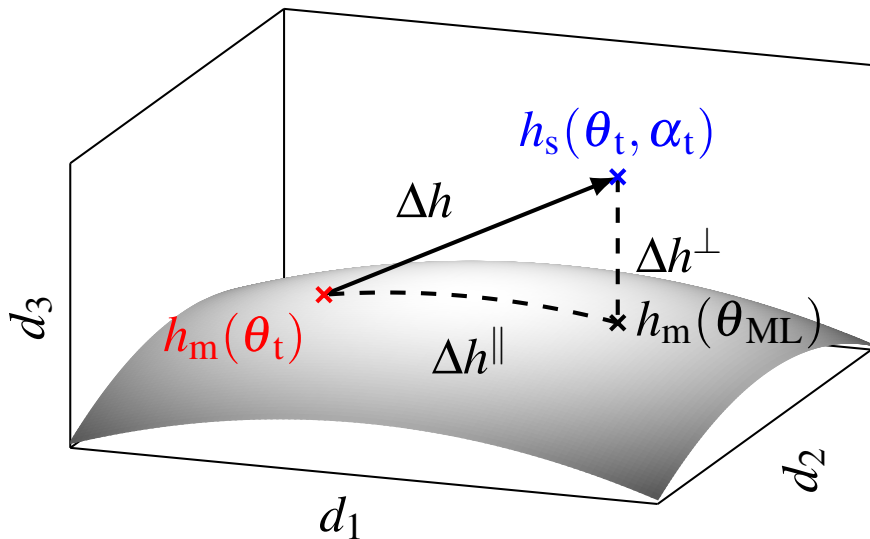
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- For example if you don't model spins and just try to measure $\mathcal{M}_c$ and q , the answers will be biased.



Perpendicular SNR and Bayes Factors

- Given an injected signal $h_{\text{sig}}(\theta_{\text{tr}}, \alpha_{\text{tr}})$, we compute the evidence for both GR and bGR [$p(\text{GR}|d)$ and $p(\text{bGR}|d)$]
- The Bayes factor compares the evidence for a beyond GR theory in the data

$$\mathcal{O}_{\text{GR}}^{\text{bGR}} \equiv \frac{p(\text{bGR}|d)}{p(\text{GR}|d)}$$

- The Bayes factor behaves like [[Vallisneri 2009, 2013](#)]

$$\log \mathcal{O}_{\text{GR}}^{\text{bGR}}|_{s_{\text{bGR}}} \sim \frac{1}{2}\rho_{\perp}^2 + \rho_{\perp}x + \frac{1}{2}x^2 \quad (\text{bGR injection})$$

$$\log \mathcal{O}_{\text{GR}}^{\text{bGR}}|_{s_{\text{GR}}} \sim \frac{1}{2}x^2 \quad (\text{GR injection})$$

where $\rho_{\perp} = \|s_{\text{bGR}}^{\perp}\|$ and x is random unit normal variable.

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$$\log \mathcal{O}_{\text{GR}}^{\text{ppE}} | s_{\text{bGR}} \sim \frac{1}{2} \left(\rho_{\perp}^{\text{ppE}} \right)^2 + x \rho_{\perp}^{\text{ppE}} + \frac{1}{2} x^2$$

where the captured SNR is

$$\rho_{\perp}^{\text{ppE}} = \text{FF}(\Delta h_{\text{bGR}}^{\perp}, \Delta h_{\text{ppE}}^{\perp}) \rho_{\perp}$$

- The *fitting factor* describes how much of the bGR signal is captured by the ppE model

$$\text{FF}(\Delta h_{\text{bGR}}^{\perp}, \Delta h_{\text{ppE}}^{\perp}) = \frac{\left(\Delta h_{\text{bGR}}^{\perp} | \Delta h_{\text{ppE}}^{\perp} \right)}{\| \Delta h_{\text{bGR}}^{\perp} \| \| \Delta h_{\text{ppE}}^{\perp} \|}$$

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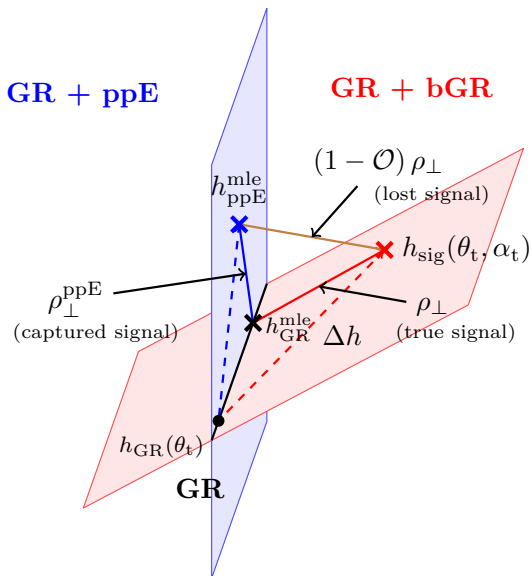
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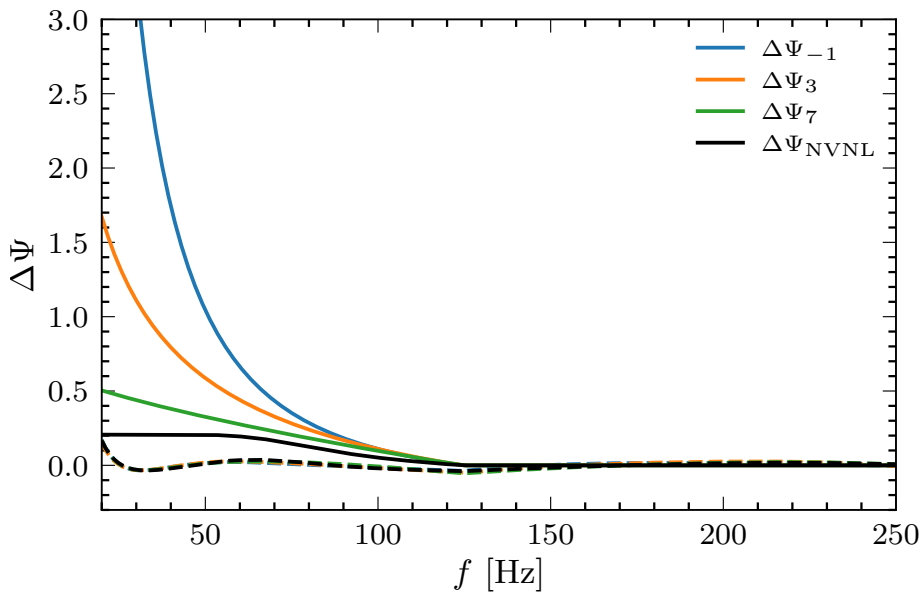
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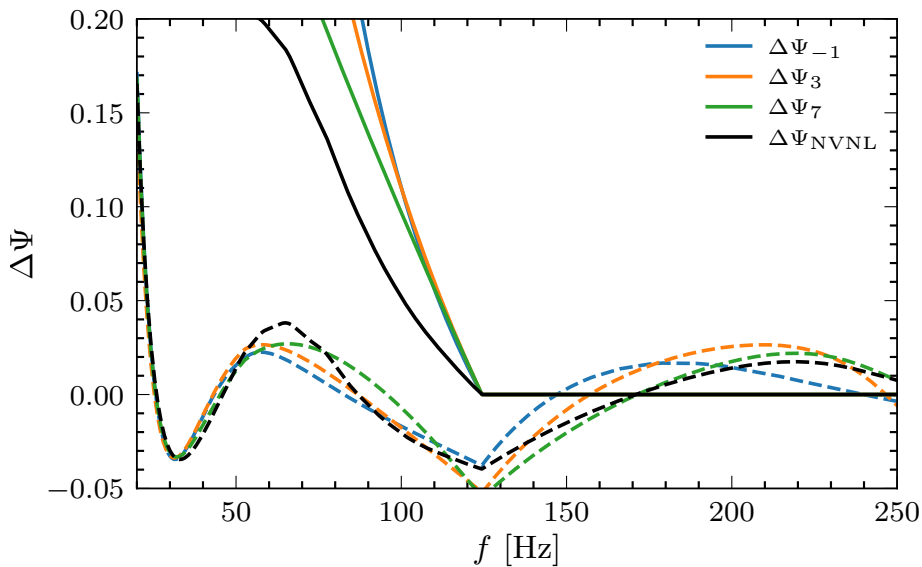
$$\text{FF}(\Delta h_{\text{bGR}}^{\perp}, \Delta h_{\text{ppE}}^{\perp}) = \frac{\left(\Delta h_{\text{bGR}}^{\perp} | \Delta h_{\text{ppE}}^{\perp} \right)}{\| \Delta h_{\text{bGR}}^{\perp} \| \| \Delta h_{\text{ppE}}^{\perp} \|}$$

- We have found that that the sensitivity loss is very small ($1 - \text{FF} \ll 1$) for some non-PN theories [Seymour+ 2024].
 - $\Delta\Psi_{\text{NVNL}} \sim e^{-f^{-1}}$ has essential singularity at $f = 0 \implies$ no ppE power

Visualization of Test

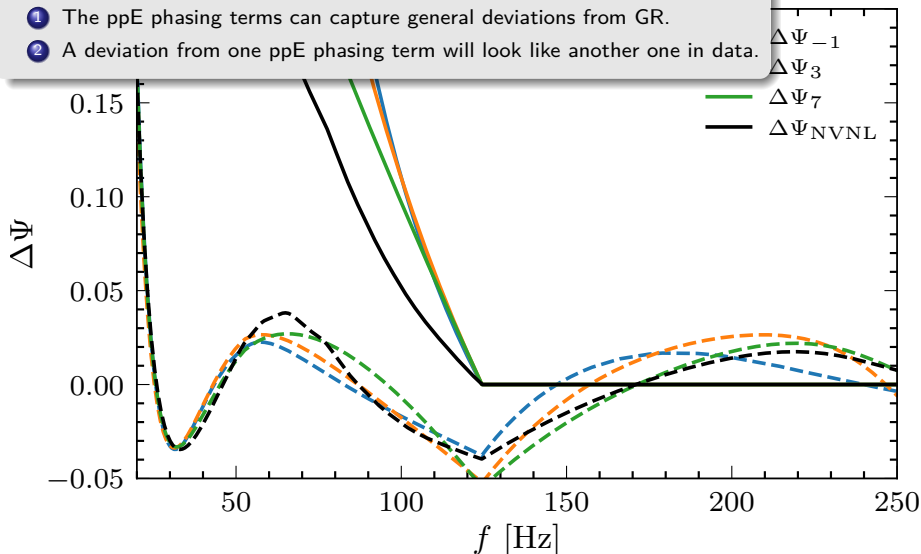




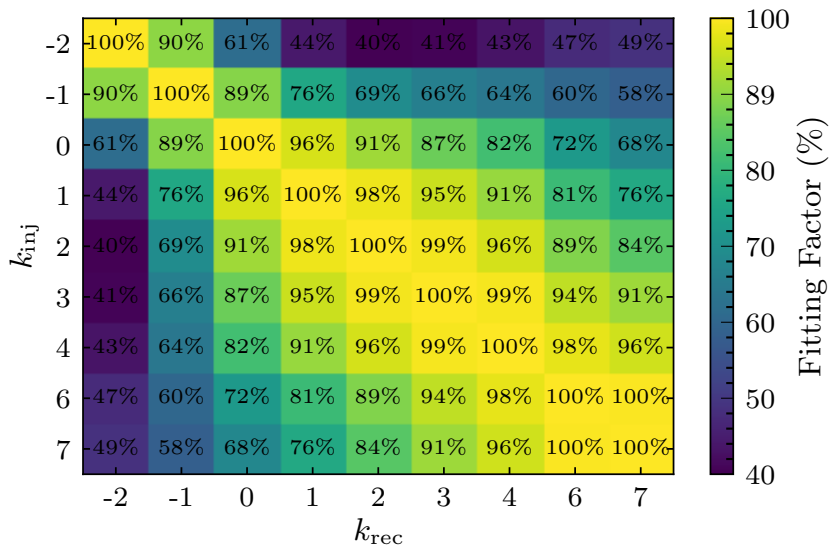


Monotonic phase deviations have some universal features in how they deviate from GR.

- ① The ppE phasing terms can capture general deviations from GR.
- ② A deviation from one ppE phasing term will look like another one in data.



Degeneracy of Multiparameter ppE



Singular Value Decomposition Approach

- Since the ppE parameter tests are degenerate, we need to identify common modes of the waveform
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$$C(\Delta h_\alpha) = \sum_a \|\Delta h_a - \mathcal{P}_n \Delta h_a\|^2$$

where the projection $\mathcal{P}_n$ projects to an orthonormal SVD basis

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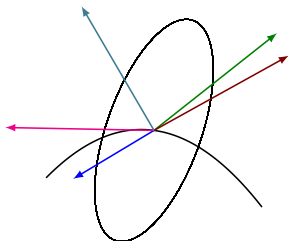
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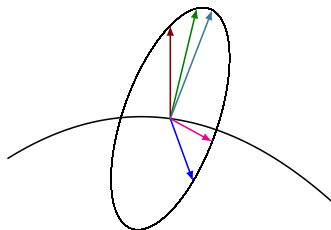
$$(\Delta h_\alpha | \Delta h_\beta) = \delta_{\alpha\beta}$$

- Goal is to find $n_{\text{svd}} \ll n_{\text{ppE}}$ but still fit signal well

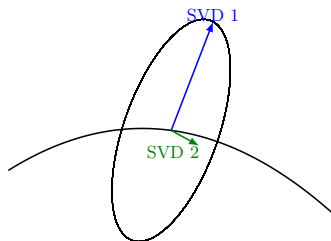
Visualization of Multiparameter SVD



Step 1: compute Δh_{ppE}

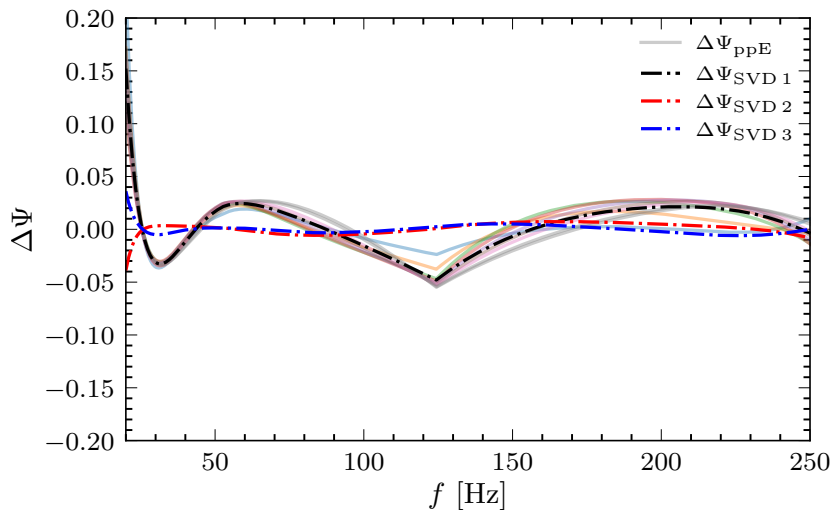


Step 2: $\Delta h_{ppE}^\perp$ computed by projecting terms parallel to $\partial_{\theta^i} h_{gr}$

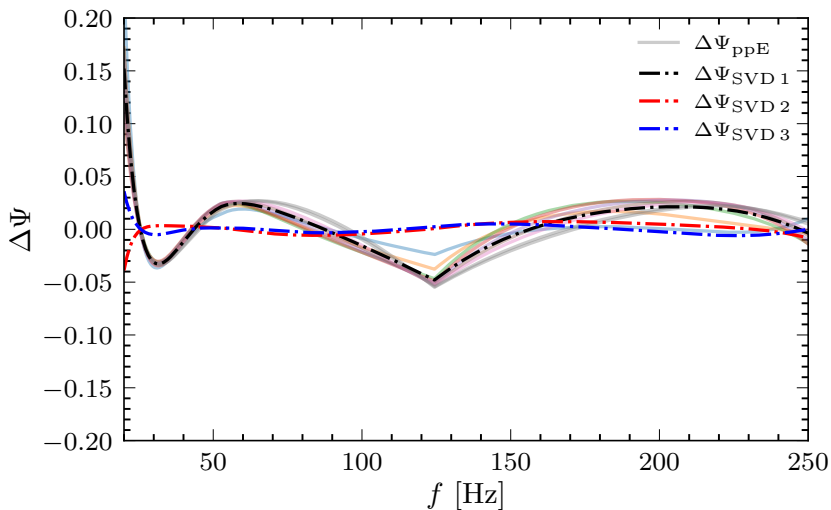


Step 3: Compute $\Delta h_{SVD}^\perp$ with the singular value decomposition

Multiparameter SVD Example for GW150914

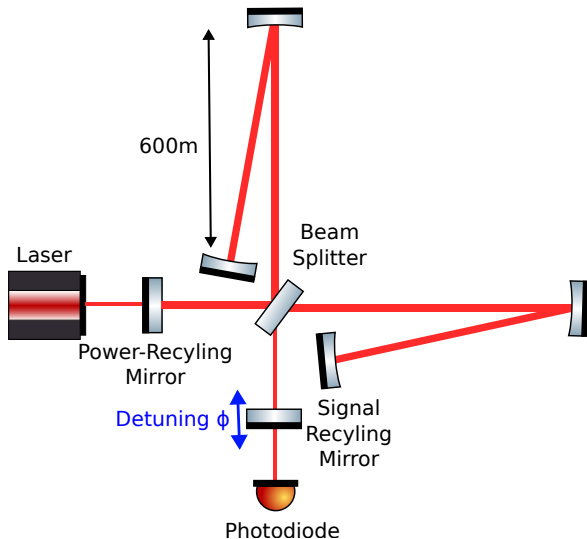


These SVD waveform modes are orthogonal so that

$$(i\Delta\Psi_{\text{SVD } a}h_{\text{gr}}|i\Delta\Psi_{\text{SVD } b}h_{\text{gr}}) = s_a^2\delta_{ab}$$


Part C : High Frequency Gravitational Wave Detection

GEO600 Optical Layout



How Detectors Measure Gravitational Waves

- The light at the photodiode is a mixture of signal and noise.

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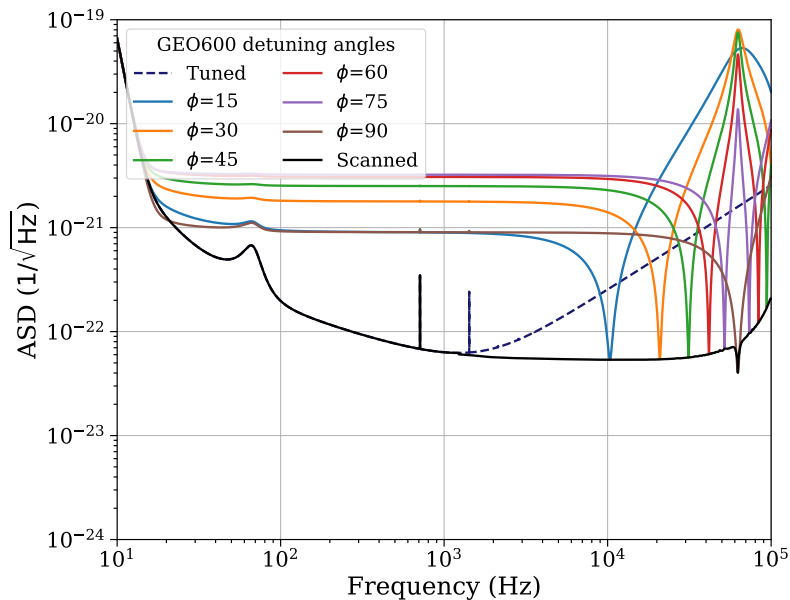
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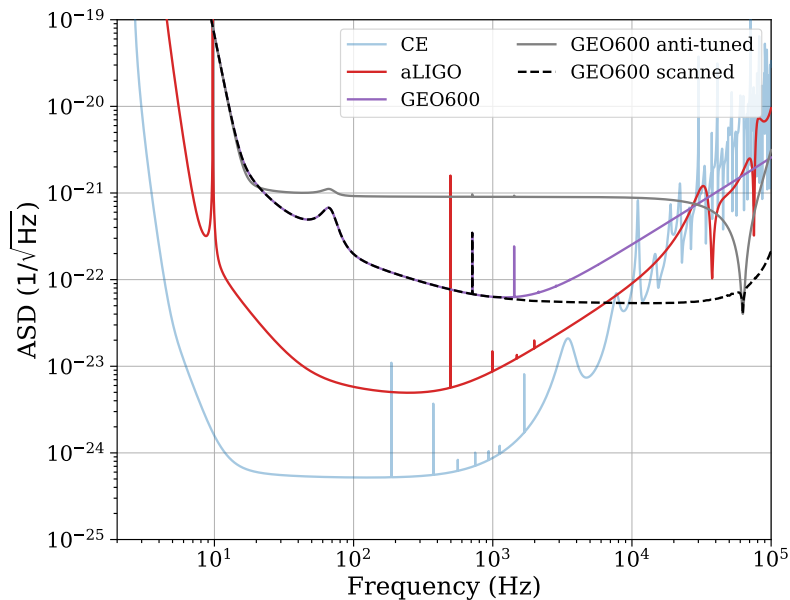
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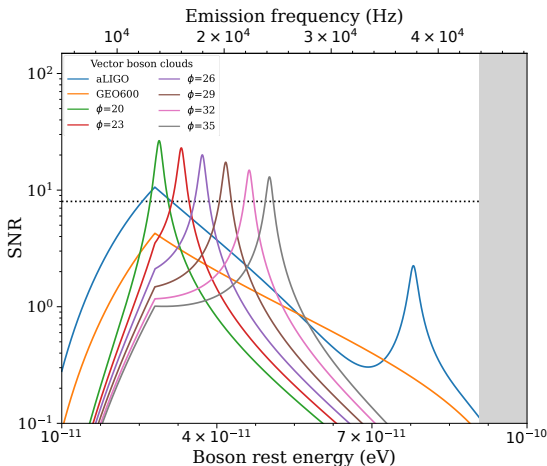
- Can we modify the **optimal frequency** that we are sensitive at by detuning the **location** of the signal recycling mirror?
- $T_h(f; \phi_{\text{srm}})$ can be adjusted to search for gravitational waves at high frequencies [Meers 1988, Mizuno+ 1993].



[backup-slide]



[backup-slide]



- GEO600 detector improves the detection prospects on boson clouds ($M = 0.3 M_{\odot}$, $\chi = 0.7$ using model from [Isi+ 2018])
- Quasircular sub-solar mass mergers $10^{-3} M_{\odot} \lesssim M \lesssim 10^{-1} M_{\odot}$ have no improvement in sensitivity.

[backup-slide]

Conclusions

- ① Studied the prospects for detecting nonviolent nonlocality in LIGO
 - Modeled the waveform in the effective-one-body formalism
 - Showed that stochastic deviations to phase are expected
 - Stacked together many events to estimate constraints in LIGO
- ② Identified geometrical meaning of tests of GR
 - Explained degeneracies and significance by geometrical framework
 - Characterized systematic error of using parameterized models
 - Used singular value decomposition to identify common modes
- ③ Modeled the effects of high frequency sensitivity in GEO600 by modulating the signal recycling cavity location.

Thanks

- Yanbei Chen for supervising this thesis
- Previous research mentors: Hang Yu, Kent Yagi, Marie Kasprzack, Arnaud Pele, Adam Mullavey, and Klebert Feitosa.
- Committee members: Katerina Chatziioannou, Saul Teukolsky, and Kathryn Zurek
- JoAnn Boyd and other TAPIR administrative staff
- NSF GRFP and other science funding bodies.

Appendix/Backup Slides

Frequency Domain Dephasing Model

- From the figure before, we saw that the random metric fluctuations produce dephasing which has a lot of structure.
- We use a principal component analysis to model the dephasing in a simple manner.

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- The mean deviation and covariance matrix are defined as

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- The principal component analysis finds optimal eigenvectors

$$\Sigma(f, f') \approx A^2 z(f) z(f'),$$

Hierarchical Tests of GR

- What we have shown is that nonviolent nonlocality predicts stochastic deviations to the phase $\Delta\Psi(f) = \zeta z(f)$.

$$\zeta \sim \mathcal{N}(0, A)$$

- This is of the same form of the hierarchical tests of GR [Isi+ 2019] which are published in the LIGO papers [LIGO+ 2021].

$$\delta\phi_k \sim \mathcal{N}(\mu_k, \sigma_k)$$

where these are the deformation coefficients.

Fisher Analysis

- We inject $\theta = (\zeta, \mathcal{M}_c, q, D_I, \iota, \psi, \alpha, \delta)$ from realistic astrophysical populations and compute the estimated variance with the Fisher matrix

$$\Gamma_{ij} = (\partial_{\theta_i} h | \partial_{\theta_j} h)$$

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where $(.|.)$ is the standard noise weighted inner product.

- From this we calculate the marginalized likelihood $p(d_a | \zeta)$ for each event a . The likelihood for the hyper parameters is

$$p(d | \mu, \sigma) = \int d\zeta p(d | \zeta) p(\zeta | \mu, \sigma)$$

- From this, we compute the posterior on the hyper parameters

$$p(\{d\} | \mu, \sigma) = \prod_a p(d_a | \mu, \sigma)$$

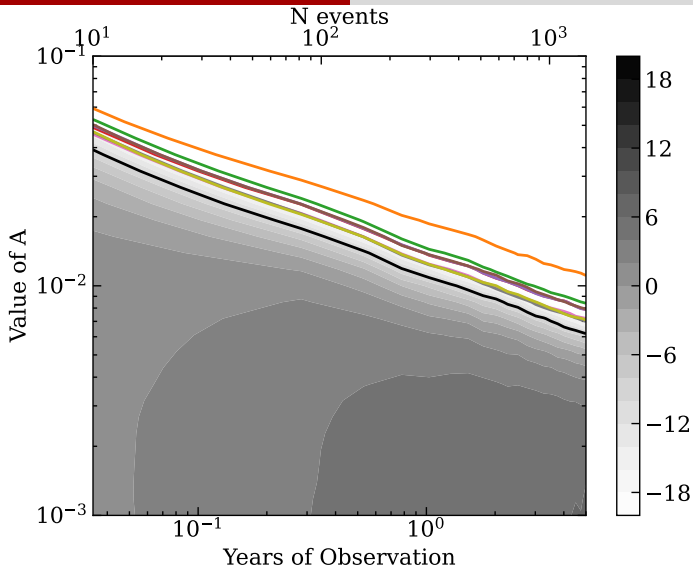
Bayes Factor

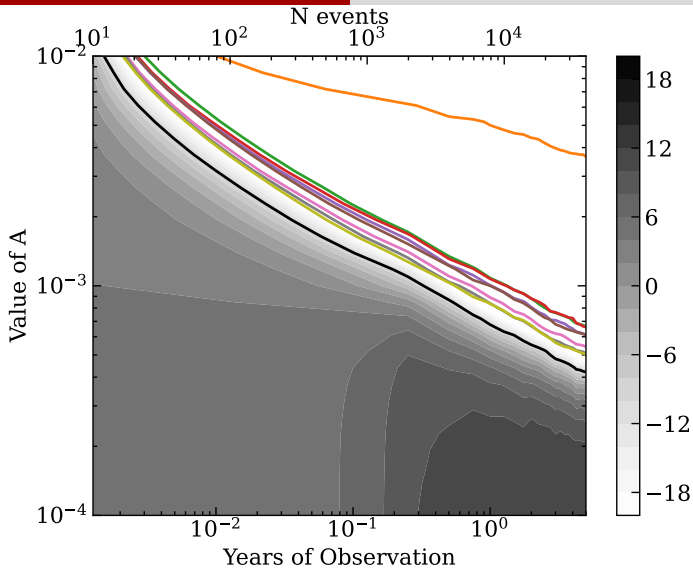
- To compare the consistency with GR, we use the log Bayes factor

$$\mathcal{B}_{\text{GR}}^{\text{NVNL}} = \log \left(\frac{p(\{d\} | M_{\text{NVNL}})}{p(\{d\} | M_{\text{GR}})} \right)$$

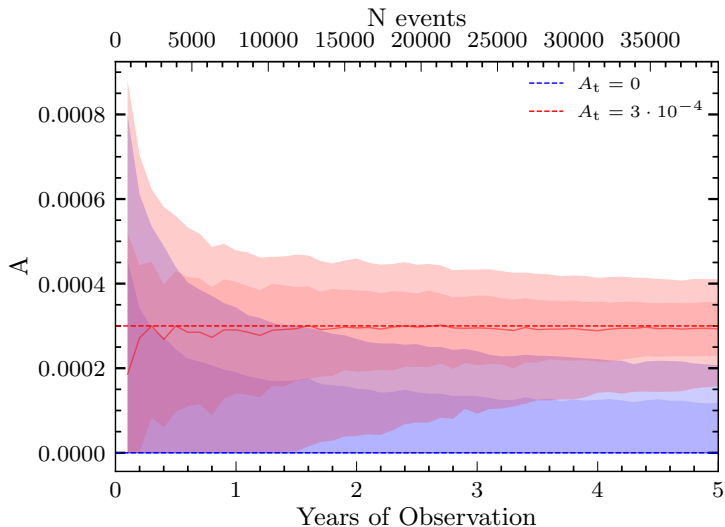
where M_{GR} and M_{NVNL} are the models.

- This is a scalar statistic which quantifies whether GR is preferred (positive) or NVNL (negative).

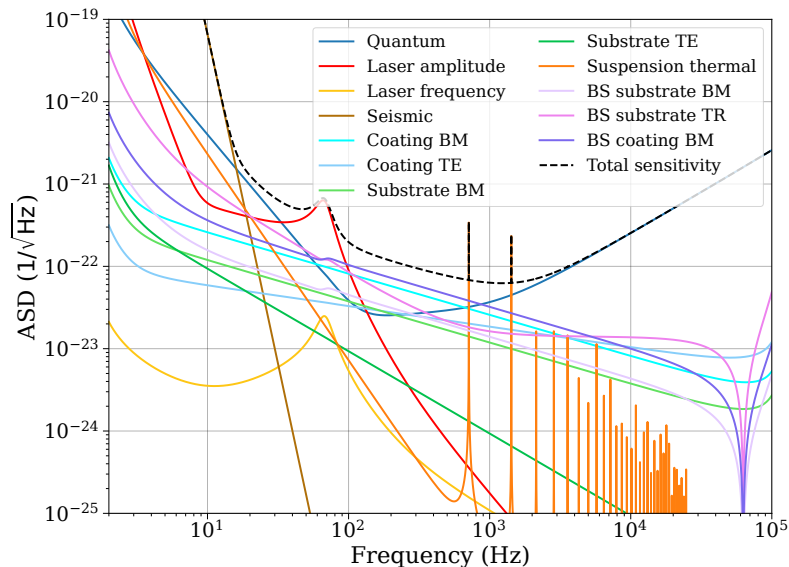




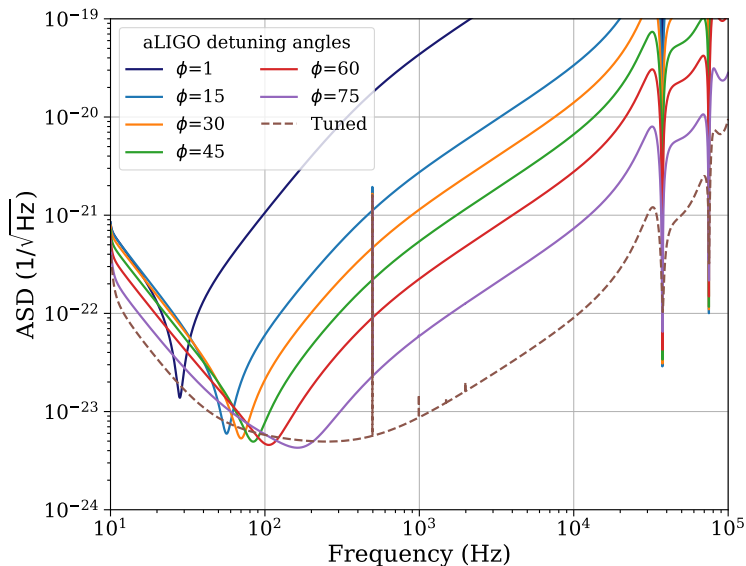
Constraints on Nonviolent Nonlocality with CE



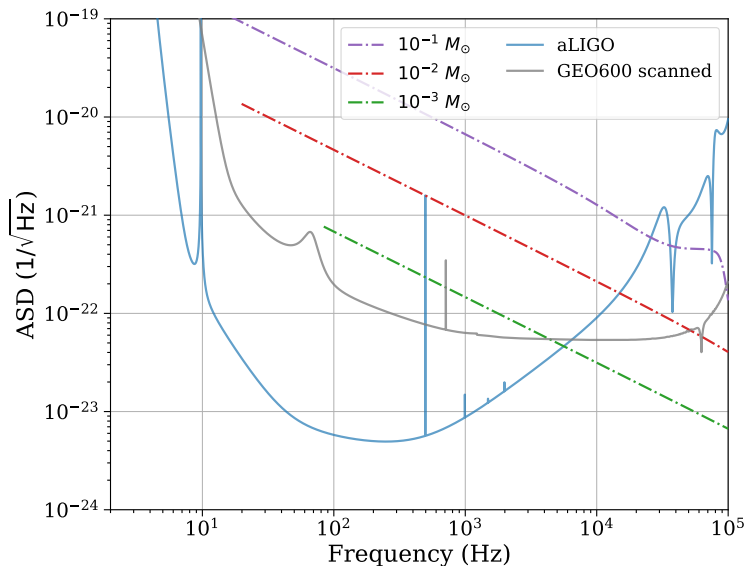
GEO600 Noise Budget



Detuning LIGO



Sub Solar Mass Detuned GEO600



Boson Clouds Full Plot

